

## ILE SURPRISE, New Caledonia

WMO: 915700

Lat: 18.480S

Lon: 163.086E

Elev: 13

StdP: 14.69

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 67.6       | 68.8 | 55.2                          | 64.9 | 71.3 | 56.9 | 68.9 | 71.8 | 27.0                  | 72.7 | 25.1 | 73.8 | 8.2                   | 150  | 0.476 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 9.9                    | 91.8            | 80.3 | 90.5 | 79.5 | 89.3 | 78.9 | 82.7                | 88.8 | 81.8 | 87.5 | 81.0 | 86.3 | 5.4                  | 110  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.0                            | 161.0 | 86.4 | 80.2 | 157.1 | 85.3 | 79.6 | 153.6 | 84.5 | 46.6          | 89.4 | 45.6 | 87.6 | 44.8 | 86.3 | 92.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 24.1          | 22.4 | 20.8 | 63.5                       | 92.8 | 3.1                | 2.4 | 61.3                                               | 94.6 | 59.5       | 96.0 | 57.8       | 97.4 | 55.6       | 99.1 |    |    |
| (5)               |      |      | 58.9                       | 83.8 | 2.1                | 1.0 | 57.4                                               | 84.5 | 56.2       | 85.1 | 55.0       | 85.7 | 53.5       | 86.4 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 79.4  | 82.8  | 83.7  | 82.8  | 81.4  | 78.9  | 76.5  | 74.8  | 74.7  | 76.2  | 78.7  | 80.4  | 82.3 | (6)  |
| (7)  |                                                                                  | DBStd       | 3.79  | 2.01  | 2.13  | 1.96  | 2.35  | 2.14  | 2.17  | 2.14  | 2.35  | 2.14  | 2.20  | 1.83  | 2.00 | (7)  |
| (8)  |                                                                                  | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD50       | 10732 | 1017  | 944   | 1017  | 941   | 897   | 794   | 769   | 764   | 787   | 889   | 912   | 1001 | (10) |
| (11) |                                                                                  | CDD65       | 5257  | 552   | 524   | 552   | 491   | 432   | 344   | 304   | 299   | 337   | 424   | 462   | 536  | (11) |
| (12) |                                                                                  | CDH74       | 43275 | 6053  | 6003  | 5965  | 4748  | 3138  | 1591  | 890   | 948   | 1569  | 2907  | 3945  | 5518 | (12) |
| (13) |                                                                                  | CDH80       | 11754 | 2042  | 2210  | 1823  | 1162  | 451   | 124   | 44    | 75    | 222   | 688   | 1066  | 1847 | (13) |
| (14) | Wind                                                                             | WSAvg       | 11.0  | 9.7   | 9.7   | 11.3  | 12.4  | 10.4  | 11.2  | 11.2  | 10.4  | 10.8  | 12.5  | 11.7  | 10.6 | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 42.2  | 5.1   | 5.4   | 7.1   | 5.1   | 3.4   | 3.0   | 2.0   | 2.0   | 1.5   | 1.6   | 2.6   | 3.5  | (15) |
| (16) |                                                                                  | PrecMax     | 66.8  | 12.4  | 14.9  | 16.1  | 15.2  | 12.9  | 8.3   | 7.0   | 5.1   | 4.7   | 4.8   | 9.8   | 12.7 | (16) |
| (17) |                                                                                  | PrecMin     | 21.2  | 0.4   | 0.5   | 1.0   | 0.4   | 0.4   | 0.3   | 0.2   | 0.4   | 0.3   | 0.2   | 0.4   | 0.4  | (17) |
| (18) |                                                                                  | PrecStd     | 11.0  | 3.2   | 3.7   | 4.1   | 3.8   | 2.5   | 2.0   | 1.5   | 1.3   | 1.2   | 1.3   | 2.0   | 3.2  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 92.7  | 94.0  | 92.4  | 94.0  | 89.0  | 85.1  | 83.0  | 84.1  | 85.5  | 88.3  | 89.9  | 91.8 | (19) |
| (20) |                                                                                  |             | MCWB  | 80.2  | 81.6  | 81.1  | 83.2  | 78.5  | 76.4  | 73.8  | 74.2  | 73.5  | 75.4  | 78.0  | 78.4 | (20) |
| (21) |                                                                                  | 2%          | DB    | 90.8  | 91.8  | 90.0  | 89.3  | 85.9  | 83.0  | 81.2  | 82.1  | 84.0  | 86.8  | 88.1  | 90.5 | (21) |
| (22) |                                                                                  |             | MCWB  | 79.4  | 81.2  | 80.2  | 79.7  | 77.3  | 75.2  | 72.4  | 72.8  | 73.1  | 75.4  | 76.8  | 78.0 | (22) |
| (23) |                                                                                  | 5%          | DB    | 89.5  | 90.2  | 88.6  | 87.2  | 84.1  | 81.4  | 79.7  | 80.4  | 82.6  | 85.5  | 86.8  | 89.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 79.0  | 80.3  | 79.8  | 79.1  | 76.4  | 74.0  | 71.5  | 71.4  | 72.5  | 74.9  | 76.1  | 77.7 | (24) |
| (25) |                                                                                  | 10%         | DB    | 88.1  | 88.7  | 87.1  | 85.6  | 82.5  | 79.8  | 78.2  | 78.7  | 81.0  | 84.0  | 85.5  | 87.9 | (25) |
| (26) |                                                                                  |             | MCWB  | 78.7  | 79.7  | 79.6  | 78.5  | 75.3  | 73.3  | 70.8  | 70.4  | 71.7  | 74.0  | 75.4  | 77.2 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 83.0  | 83.8  | 83.2  | 84.9  | 80.9  | 79.0  | 77.1  | 76.9  | 76.8  | 78.8  | 80.9  | 81.7 | (27) |
| (28) |                                                                                  |             | MCD B | 89.0  | 90.9  | 89.6  | 91.5  | 85.7  | 82.7  | 80.1  | 81.3  | 82.4  | 85.1  | 86.8  | 88.5 | (28) |
| (29) |                                                                                  | 2%          | WB    | 81.8  | 82.8  | 82.0  | 81.5  | 79.1  | 77.4  | 75.5  | 75.3  | 75.4  | 77.4  | 79.1  | 80.3 | (29) |
| (30) |                                                                                  |             | MCD B | 87.7  | 89.3  | 87.2  | 86.9  | 83.3  | 80.3  | 78.3  | 79.2  | 81.1  | 83.7  | 84.9  | 86.8 | (30) |
| (31) |                                                                                  | 5%          | WB    | 80.8  | 81.8  | 81.3  | 80.4  | 78.0  | 76.2  | 74.1  | 73.9  | 74.3  | 76.4  | 78.0  | 79.4 | (31) |
| (32) |                                                                                  |             | MCD B | 86.5  | 87.8  | 86.1  | 85.3  | 81.9  | 79.3  | 77.3  | 77.7  | 79.6  | 82.5  | 83.6  | 85.8 | (32) |
| (33) |                                                                                  | 10%         | WB    | 79.9  | 80.9  | 80.5  | 79.3  | 77.1  | 75.1  | 72.8  | 72.8  | 73.2  | 75.4  | 76.9  | 78.6 | (33) |
| (34) |                                                                                  |             | MCD B | 85.3  | 86.4  | 85.0  | 83.6  | 80.7  | 78.6  | 76.6  | 76.8  | 78.2  | 81.4  | 82.4  | 84.8 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 10.1  | 9.9   | 8.6   | 8.2   | 7.3   | 7.0   | 7.3   | 8.4   | 10.0  | 10.6  | 10.7  | 11.0 | (35) |
| (36) |                                                                                  |             | MCDBR | 12.6  | 12.1  | 10.7  | 11.0  | 9.6   | 8.5   | 9.0   | 10.2  | 11.9  | 12.3  | 12.5  | 12.8 | (36) |
| (37) |                                                                                  |             | MCWBR | 5.0   | 4.8   | 4.7   | 5.4   | 5.0   | 4.6   | 4.9   | 5.4   | 5.6   | 5.1   | 5.1   | 5.0  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 10.4  | 10.6  | 9.1   | 9.5   | 8.3   | 7.3   | 7.6   | 8.5   | 10.2  | 10.6  | 10.9  | 10.9 | (38) |
| (39) |                                                                                  |             | MCWBR | 5.0   | 4.8   | 4.6   | 5.2   | 4.7   | 4.5   | 4.9   | 5.0   | 5.2   | 4.9   | 5.1   | 4.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.391 | 0.390 | 0.389 | 0.382 | 0.373 | 0.373 | 0.363 | 0.361 | 0.369 | 0.385 | 0.389 | 0.381 | (40) |      |
| (41) |                                                                                  | taud        | 2.595 | 2.605 | 2.616 | 2.618 | 2.607 | 2.600 | 2.596 | 2.601 | 2.555 | 2.538 | 2.551 | 2.604 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 303   | 299   | 292   | 283   | 274   | 268   | 274   | 285   | 294   | 298   | 301   | 306   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 33    | 33    | 31    | 29    | 28    | 27    | 28    | 29    | 33    | 35    | 35    | 33    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2128  | 2059  | 1755  | 1540  | 1301  | 1197  | 1270  | 1522  | 1855  | 2052  | 2114  | 2220  | (44) |      |
| (45) |                                                                                  | RadStd      | 195   | 180   | 157   | 154   | 98    | 69    | 89    | 92    | 125   | 141   | 128   | 182   | (45) |      |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page